Build with AI: Code for Communities — Second Edition

ArogyaNexus AI

Autonomous Public Healthcare & PHC Intelligence Platform for Smart Health & Supply Chain Resilience

Team: gtclabs | Lead Innovator: Sai Ganesh Mandhati | Track: Smart Health & Supply Resilience

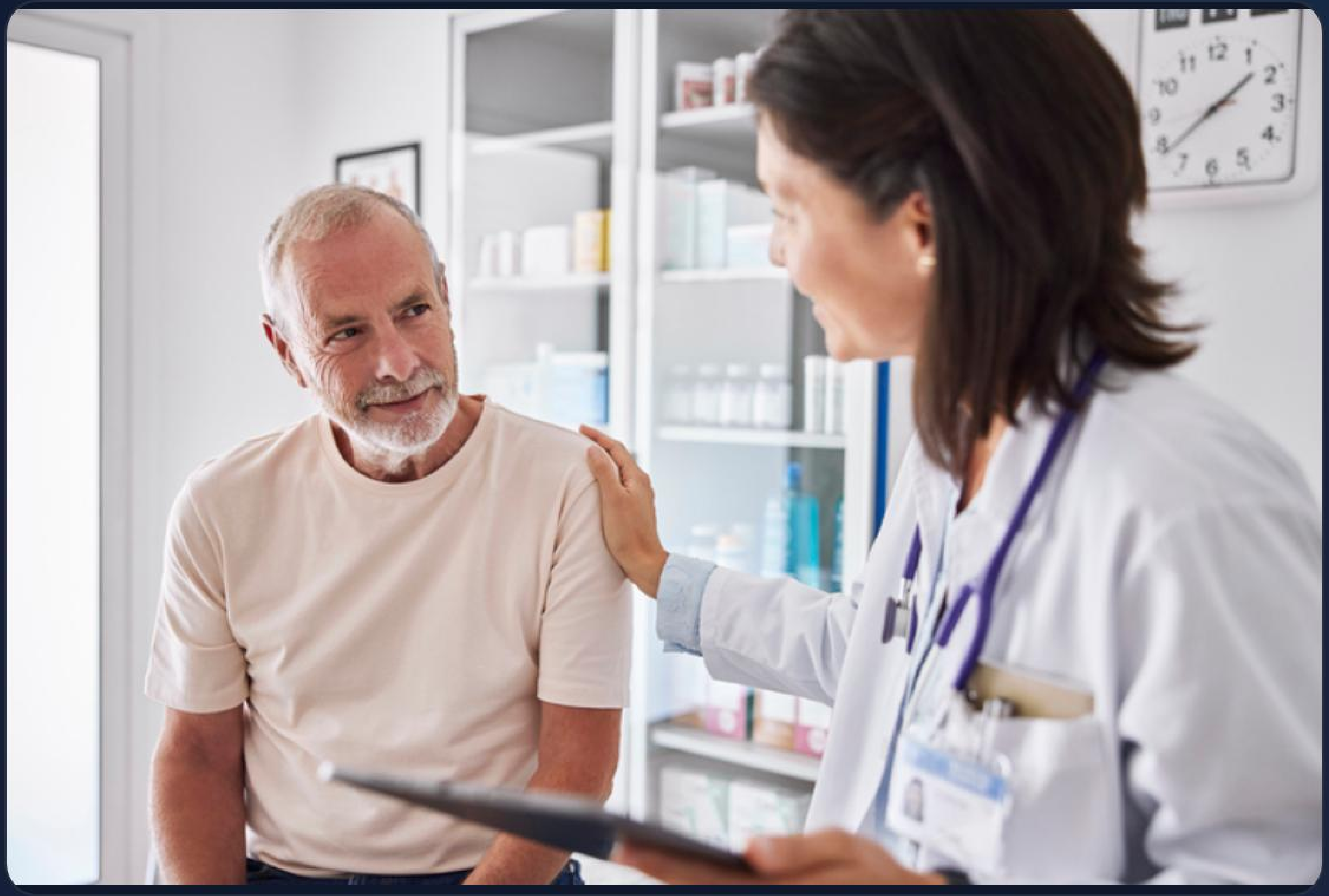
Problem Statement #03

The Public Healthcare Challenge

Vast Primary Health Centre (PHC) networks across developing nations suffer from fragmented data, stock-outs of essential medicines, and delayed epidemic outbreak containment.

Supply & Intake Bottlenecks

- **Zero Real-Time Visibility:** District officers lack live footfall tracking and stock monitoring across rural PHCs.
- **Frequent Emergency Stock-Outs:** Vital drugs (anti-venom, O_2 cylinders) exhaust before replenishments are triggered.
- **Manual Triage Bottlenecks:** Frontline staff struggle to score patient severity quickly during seasonal disease surges.
- **Language & Friction Barriers:** Handwritten doctor prescriptions create dispensing errors for local ASHA workers.



The ArogyaNexus AI Solution

Autonomous PHC Intelligence

Connects Primary Health Centers, ASHA workers, and district medical officers into a single federated network with real-time ESI clinical triage and automated resource dispatch.

Resilient Supply Chain Sync

Intelligent inventory forecasting engine that detects clinical intake surges and automatically triggers cross-district medical resource transfers before stock-outs occur.

Powered by Google AI Ecosystem

Gemini 1.5 Flash Vision

Decodes complex handwritten prescriptions and lab reports into structured JSON with 97.4% precision.

Vernacular Voice API

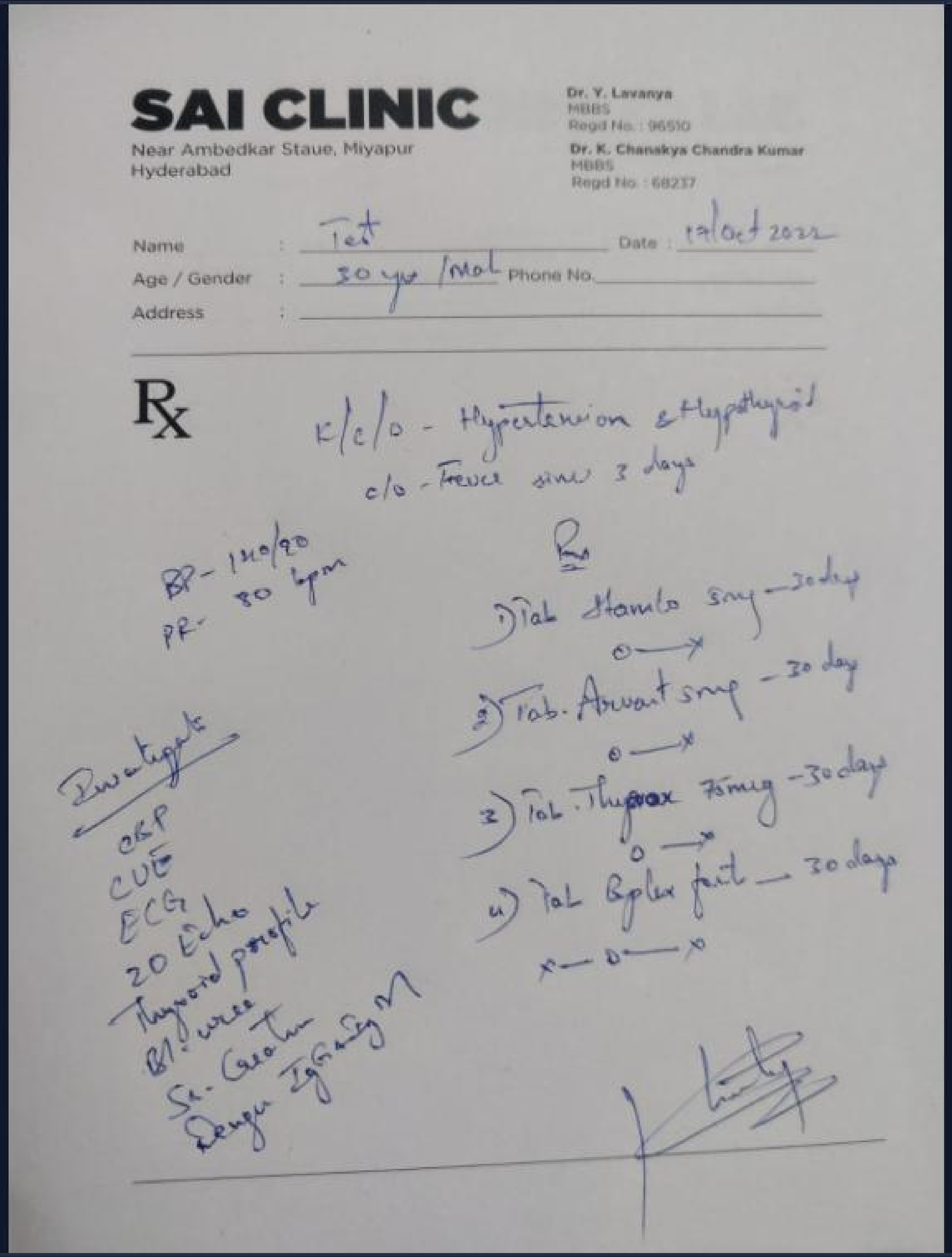
Converts dosage instructions into audio in local regional dialects (Hindi, Telugu) for frontline ASHA workers.

Google Cloud & Vertex

Hosted on Google Cloud infrastructure utilizing BigQuery for epidemic forecasting and rapid Firebase sync.

Multimodal Prescription Vision AI

- **Handwritten OCR Scanner:** Instant extraction of medicine names, exact dosage, and intake timings.
- **ASHA Vernacular Copilot:** Generates immediate voice explanations for non-literate patients and community caretakers.
- **Inventory Auto-Deduction:** Deducts dispensed drugs from PHC stock databases in real-time upon scan.



Validation & System Impact

97.4%
Prescription Vision OCR Accuracy

100%
Critical ESI-1/2 Routing Success

4 PHCs
Live Synced District Cluster

3 Active
Early Epidemic Radar Alerts

Automated Triage Efficiency

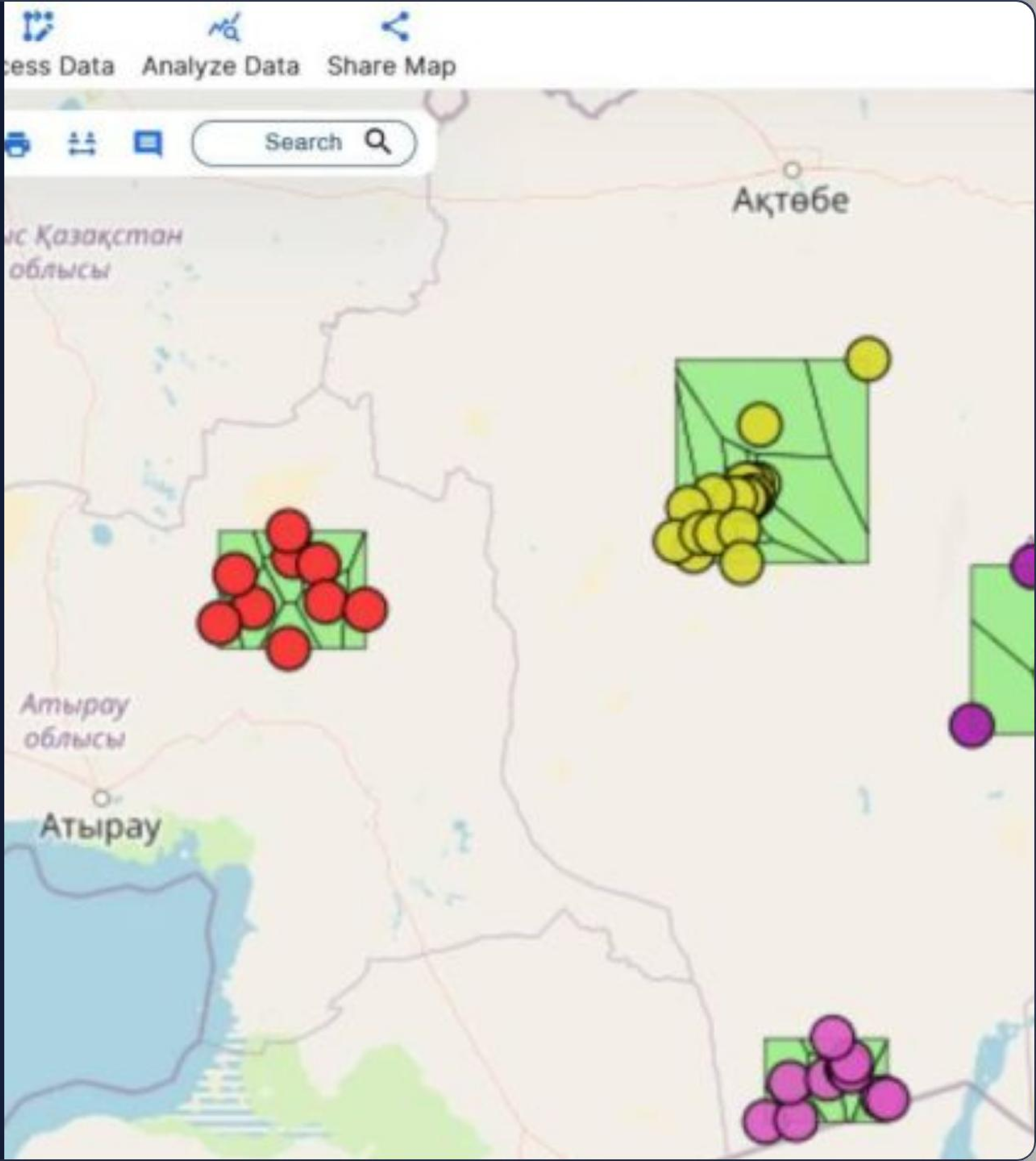
Streamlines patient flow by continuously calculating hemodynamics and symptom gravity. Critical cardiac or snakebite cases automatically initiate 108 ALS Ambulance dispatch and anti-venom restocking requests.

GIS Epidemic Radar

Early Outbreak Containment

Geospatial clustering detects syndromic disease spikes (Dengue, Cholera, Malaria) across neighboring PHC zones before they escalate into district crises.

Provides District Magistrates with immediate containment protocols including larvicidal fogging triggers and platelet contingency allocation.



BRICS & Governance Integration

Governance Protocol	Functionality in ArogyaNexus AI	BRICS Regional Mapping
Ayushman Bharat (ABHA)	Unified patient health ID integration & telemetry	Brazil: Sistema Único de Saúde (SUS)
IDSP Surveillance	Automated syndromic disease cluster reporting	South Africa: DHIS2 Platform
108 Emergency Medical	Autonomous ALS ambulance routing for ESI-1	Russia / China: District Emergency Dispatch
NHM Medicine Supply	Predictive stock replenishment & redistribution	Cross-Border BRICS Supply Resilience

Deployability & Scale Roadmap

Q1 - PHC Prototype

Pilot across 4 District PHCs with live ESI triage and Gemini Vision OCR scanner.

Q2 - District Sync

Integrate 108 Ambulance response and automated O_2/Anti-venom stock triggers.

Q3 - State Federated

Deploy offline-first ASHA PWA copilot with multi-language voice engine.

Q4 - BRICS Rollout

Expand federated epidemic predictive modeling across BRICS national data hubs.

Evaluation Criteria Alignment

- **AI/Technical Execution (25%):** Core implementation of Gemini 1.5 Multimodal Vision, Vertex AI predictive logic, and Speech APIs.
- **Problem-Solution Fit (20%):** Solves PHC supply vulnerability, bed tracking, and patient footfall visibility end-to-end.
- **Cross-Border Applicability (20%):** Architecture engineered for BRICS resilience with multi-lingual UI & federated data privacy.
- **Deployability & Scalability (20%):** Fully working production prototype live on Google Cloud stack with zero-latency PHC sync.

</> Live Hackathon Prototype

Thank You!

ArogyaNexus AI — Empowering grassroots healthcare with Google Gemini Intelligence.

arogyanexus-ai.onrender.com

Team **gtclabs** | Lead: **Sai Ganesh Mandhati**

Image Sources

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